



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Status Update 4

Group #17: Patent Miner

List of team members:

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TA: Sabyasachi Gupta

Sponsor: Dr. Reddy Narasimha



Project Summary

Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.

The image shows two side-by-side screenshots of a patent database interface. The left screenshot displays a list of search results for the keyword 'localization'. The right screenshot shows a detailed view of a specific patent document.

Search Results Table:

Date Publish...	Family ID	Pages	Title
25-09-25	10000078...	28	SCOPE TURRET
25-09-25	10000086...	91	COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER
25-09-25	10000086...	70	SOY-DERIVED BIOACTIVE PEPTIDES FOR USE IN COMPOSITIONS AND METHODS
25-09-25	10000086...	12	APPARATUS, SYSTEM, AND METHOD OF PROVIDING HAZARD DETECTION
25-09-25	10000086...	134	Interconnectable Tiling System
25-09-25	10000086...	40	Smart specialized brushing device and Systems and Methods
25-09-25	10000086...	22	BIOMETRIC AUTHENTICATION
25-09-25	10000086...	16	PERSONAL AUDIO ASSISTANT DEVICE AND METHOD
25-09-25	10000086...	64	SYSTEM AND METHOD TO HELP ENABLE CREATION OF, DISTRIBUTION OF, AND USE OF
25-09-25	10000086...	42	Organic-Inorganic Hybrid Bulk Assemblies and Methods
25-09-25	10000086...	10	SYSTEM AND METHOD FOR WATER TREATMENT INCENTIVE
25-09-25	62242490	24	Method and system for integratedly managing vehicle operation state
25-09-25	77859897	63	PEPTIDE
25-09-25	78134919	88	TREATMENT INVOLVING NON-IMMUNOGENIC RNA FOR ANTIGEN VACCINATION
25-09-25	78413837	30	METHOD FOR PRODUCING MILK LIKE PRODUCTS
25-09-25	78598778	648	GASTRIC INHIBITORY PEPTIDE RECEPTOR LIGANDS
25-09-25	80218437	95	LIVE ATTENUATED SARS-COV-2 AND A VACCINE MADE THEREOF

Document Viewer Details:

COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER

DOCUMENT ID	DATE PUBLISHED
US 20250297317 A1	2025-09-25

INVENTOR INFORMATION

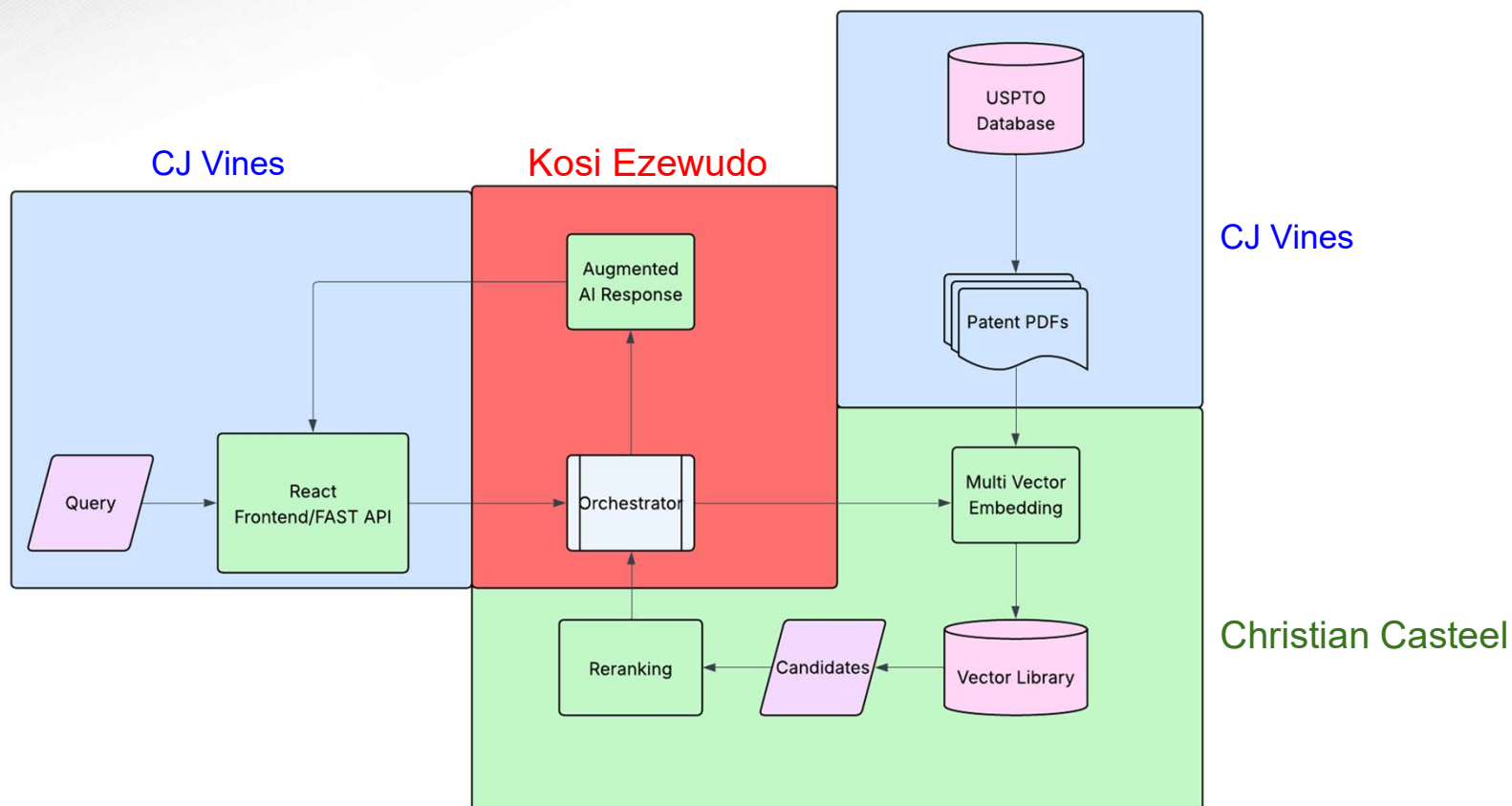
NAME	CITY	STATE	ZIP CODE	COUNTRY
Sedlak; Joseph Charles	Waltham	MA	N/A	US
Winn-Deen; Emily	Waltham	MA	N/A	US
Gusenleitner; Daniel	Waltham	MA	N/A	US
Couvillon; Anthony David	Waltham	MA	N/A	US
Bortolin; Laura Teresa	Waltham	MA	N/A	US
Salem; Deniel Parker	Waltham	MA	N/A	US
Biette; Kelly	Waltham	MA	N/A	US

APPLICATION NO 18/580422
DATE FILED 2022-07-21

US CLASS CURRENT: 1/1

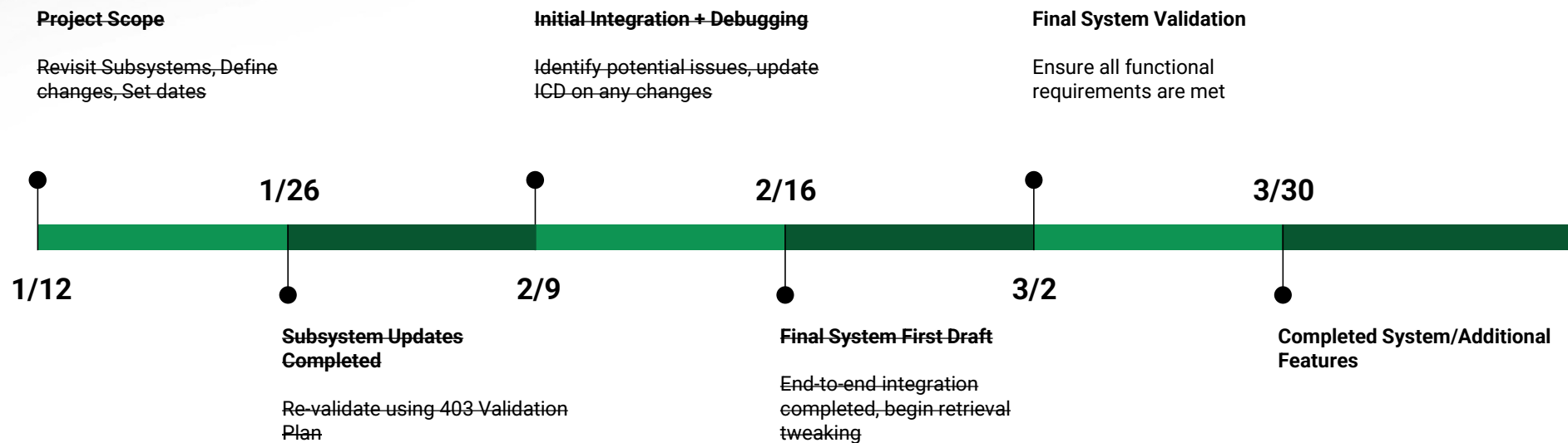


System Overview





Project Timeline





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The Patent Miner

Rusty uncovers innovation, one swing at a time 🛠️

K-5

Search



Embedding Pipeline and Storage

Christian Casteel

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Hand labelled new validation query set based upon new embedding pipeline • Ran preliminary analysis on quantized and dimensionality reduced validation data	• Allow parameterization and testing from the frontend • Complete analysis of new validation data • Create final “per patent” validation



Embedding Pipeline and Storage

Christian Casteel

Claim: A capacitive strain sensor configured to measure a strain, the sensor comprising:
a dielectric layer;
a first electrode placed on a first face of the dielectric layer...

Query 1: Capacitive strain sensor with fragmented carbon nanotube electrode layer producing strain-dependent electrical response

Query 2: A deformation sensor that increases electrical response by using a crack-forming conductive nanostructured layer that changes conduction as stretching occurs

Metrics with old validation set

run	precision@10	recall@10	ndcg@10	mrr@10	candidate_hit@400
bm25	0.09	0.9	0.643757644	0.559444444	1
vector	0.0325	0.325	0.27693311	0.261458333	0.325
hybrid_a02	0.09	0.9	0.643757644	0.559444444	1
hybrid_a05	0.09	0.9	0.644853238	0.560694444	1
hybrid_a08	0.075	0.75	0.579690469	0.523571429	0.975

Metrics with current validation set

run	precision@10	recall@10	ndcg@10	mrr@10	candidate_hit@400
bm25	0.162	0.309820728	0.311311773	0.512333333	0.92
vector	0.06	0.10762465	0.145510008	0.331190476	0.46
hybrid_a02	0.164	0.316487395	0.3192765	0.529	0.92
hybrid_a05	0.162	0.311487395	0.314257033	0.517666667	0.86
hybrid_a08	0.114	0.212848739	0.265385045	0.521190476	0.84



Frontend/Preprocessing CJ Vines

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Embedded 50k patents • Dimensionality reduction success! • Implemented better testing methodology, parameter sweep for comparison of different alpha values, limits, @k amounts, etc.	• Source cloud services (TAMU workstation? Weaviate paid subscription?) • Scale up as high as possible on home PC • Implement Metadata Filtering



Weaviate Dilemma Pt. 2

- Dimensionality reduction results indicate that we can reduce space by about 6 times at the cost of accuracy
 - Dr. Reddy appears to care a lot about scalability
 - MB25 solves some retrieval issues
- We can dump LMDB and just use Weaviate export
 - Slight latency cost
 - Doesn't really matter for a pre-production POC



Weaviate



Retrieval Augmented Generation (RAG) Subsystem

Kosi Ezewudo

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Full system refinement: ○ Prompt engineering ○ Removal of Metadata extraction Layer ○ Implementation of memory	• Continue Full system and Sub-system refinement ○ Finalization of memory implementation ○ Reference retrieved context in response • Brainstorm potential pathways for; ○ Implementation of claims defense/combat dedicated layer. ○ Visualization of retrieved patents and prior art • Perform extensive validation

Retrieval Augmented Generation Subsystem Kosi Ezewudo

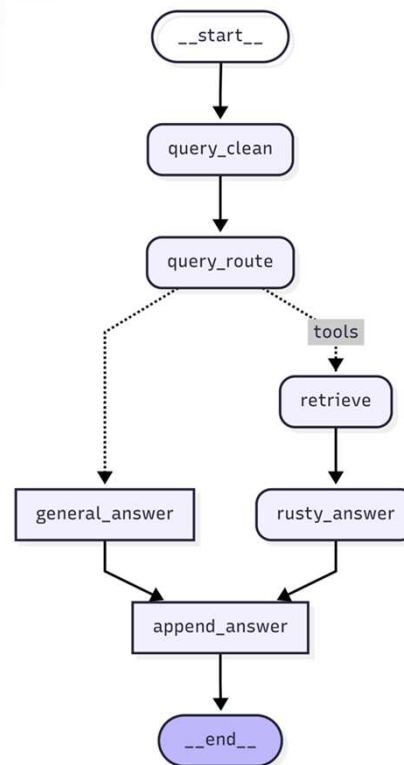


Fig 1. Updated Backend Pipeline

Execution Plan

[illegible]



Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Reranking Test	"Golden" query - chunk pairs must have $\geq .05$ Recall@k, Mean Reciprocal Rank (MRR) $\geq .5$, and Normalized Discounted Cumulative Gain (NDCG) of .5	k = 10 target chunks at minimum with an equal number of matching queries. Calculate Recall@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents.	Untested	Christian Casteel
3.2.1.5.	Embedding Accuracy Test	For a set of chunks, Hit@k of 1 and precision at k above .5. Relevance determined by patent id	Query containing information from multiple patents with similar CPC codes will test retrieval. Calculate Hit@k and Precision@k on top k chunks where k = 10.	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty sections and incorrect CPC codes will be sent through the pipeline.	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via frontend download script. The bulk dataset should contain minimum 1000 patents.	Untested	Christian Casteel & CJ Vines
3.2.1.6.	Query Metadata Extraction	Extract all relevant metadata from user query, and use the metadata to filter patent search	Execute patent search using queries containing various metadata including prior date, patent ID, CPC etc.	Untested	Kosi Ezewudo
3.2.1.3.	Query Cleaning	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Accuracy and Quality Test	Generated responses must have $\geq .8$ Answer relevancy scores, $\geq .8$ Diversity and Novelty	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Diversity and Novelty scores	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines



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Thank you!